TEAM T070

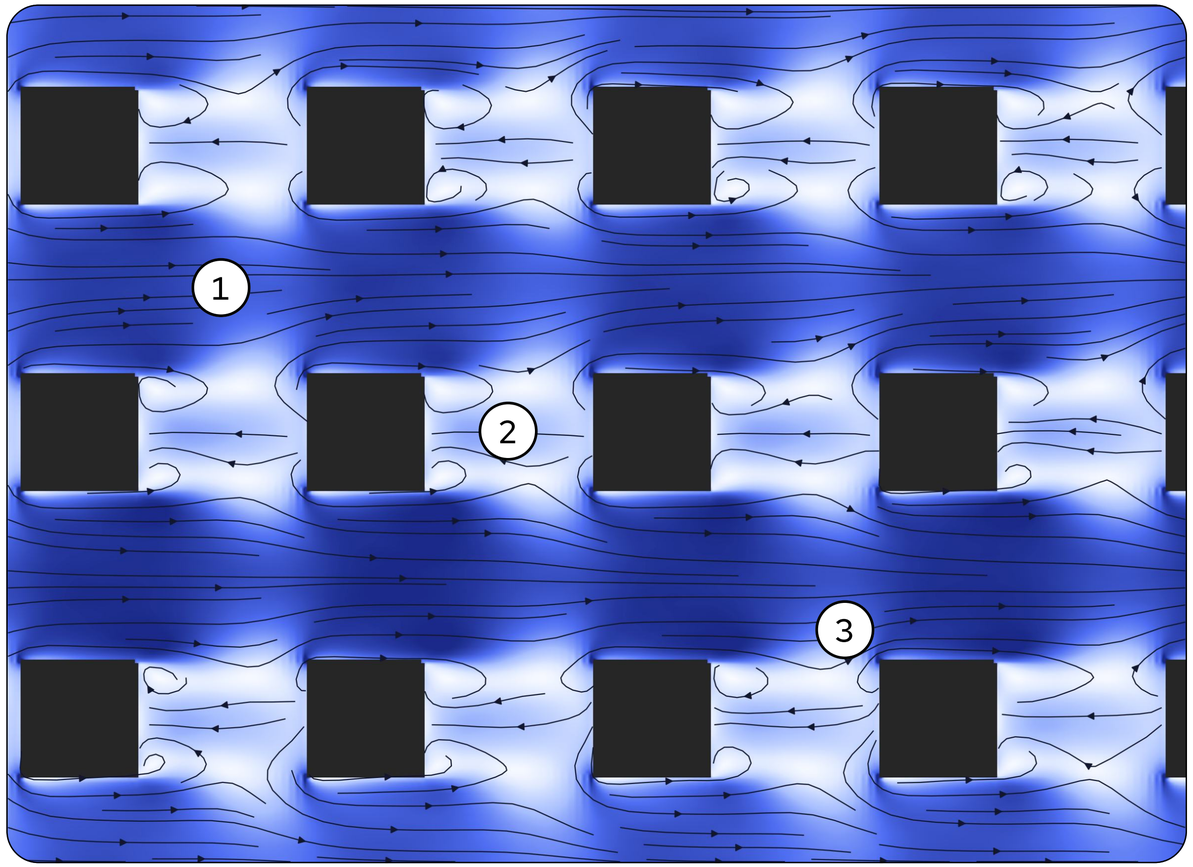
Urban Microclimate Modelling

Machine-learning surrogates for pedestrian-level wind, trained on 538 large-eddy simulations

Ayush Aryan, Aparna Singh, Mahir Abdullah, Om Mishra, Neeraj Singh

Advanced Machine Learning, Phase 1

Buildings reshape the wind at street level



slow fast

Real simulation (case UA16): wind 1.75 m above the ground, blowing from the left. Lines follow the average wind.

1. **Jet**
   air speeds up along the open street
2. **Wake and eddies**
   slow, swirling air behind each building
3. **Deflection**
   the wind bends around each building

This street-level wind decides pedestrian comfort, safety in gusts, and how well streets ventilate.

Simulating it costs thousands of CPU-hours per neighbourhood: too slow to test many layouts.

A surrogate model learns from finished simulations and approximates them, so a new layout needs no new simulation.

Can a model predict the wind at street level from the building layout alone?

Using only the building geometry and the wind direction, and tested on cities and layouts the model has never seen.

What goes in, and what comes out

INPUT

Building-height map

the height of every 1 m cell



Wind direction θ

the only weather input that changes between cases



MODEL

Surrogate model

learned once from finished simulations, then reused for any new layout

OUTPUT

TASK A

One number per neighbourhood

the mean wind over open ground, written U_mean / u*

TASK B

The full wind map

U / u* at every open cell, 1.75 m above the ground

Success means accurate predictions for cities and layouts the model has never seen.

UrbanTALES: 538 simulations of wind in cities

224

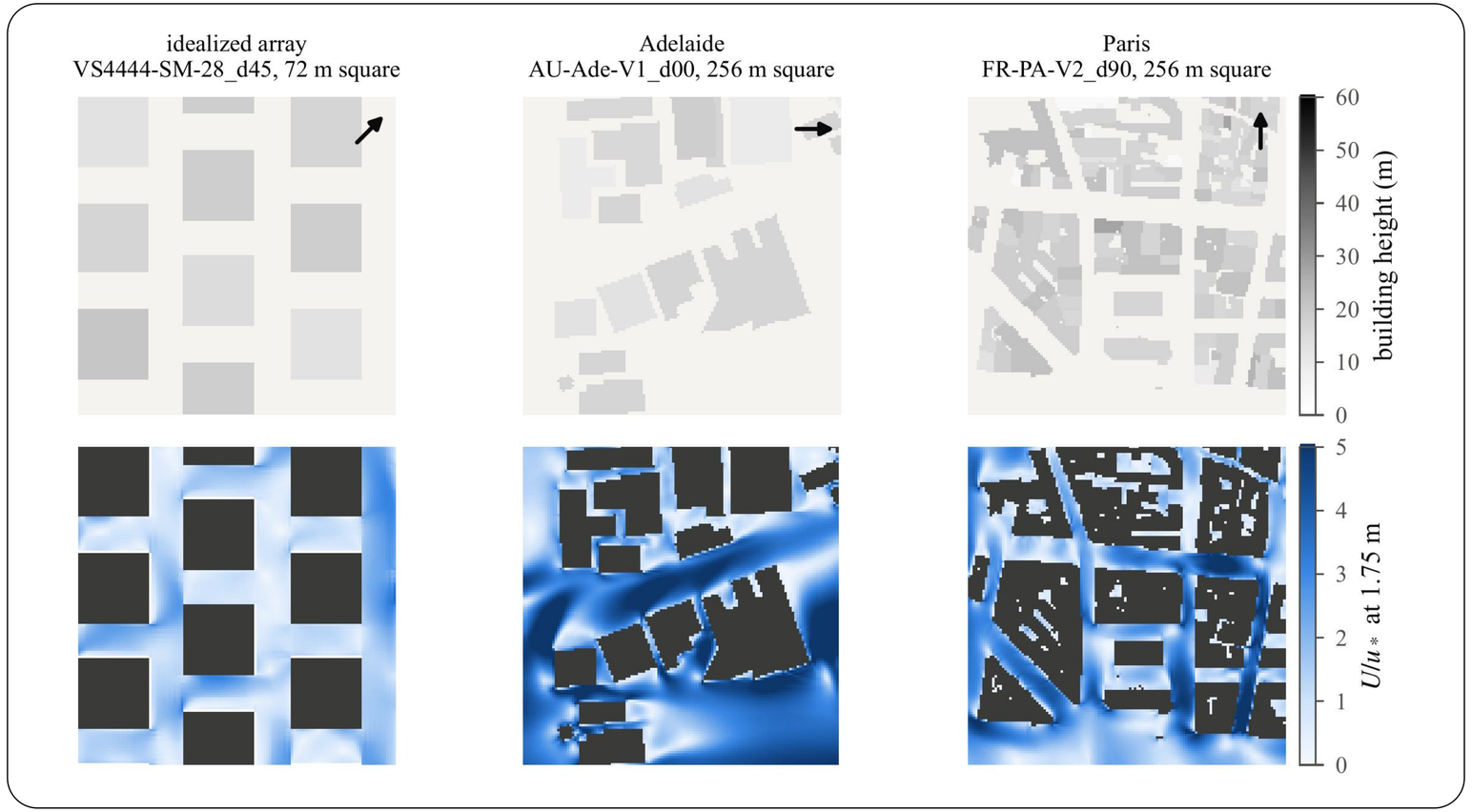
idealized block arrays: 8 building densities, 2 wind directions

314

real neighbourhoods in 22 countries, built from OpenStreetMap

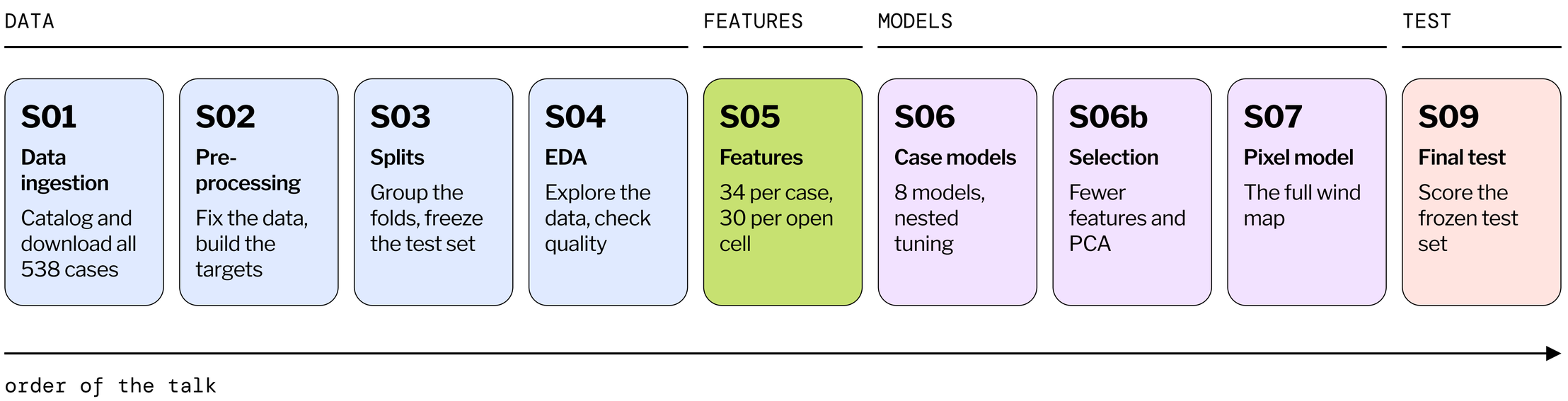
1.75 m

pedestrian height, where the time-averaged wind is recorded



Top: building heights. Bottom: simulated wind at 1.75 m. Jets form along open streets, wakes behind buildings.

Nine stages, presented in the order we built them



Each stage reads the outputs of the stage before it. The repository has no S08: the frozen test set is scored in S09.

S01: all 538 cases, downloaded and checked

Building raster

`<case>_topo`

the height of every cell; 0 means open ground

Pedestrian-level field

`<case>_ped.nc`

time-averaged wind and turbulence at 1.75 m

Vertical profiles

`profile-<case>.csv`

wind averaged over the area, at each height

Run configuration

`<case>_p3d`

the settings, including the pressure gradient that sets the wind direction

538

cases in data/catalog.csv, one row each with its file links and its group

4.8 GB

downloaded, of 6.05 GB in total: only the four file types we need

Restartable

parallel download with retries and a socket timeout, so a stalled file never hangs a worker

Every file is checked after download. The problems we found inside the files are fixed in S02.

S02: three data problems, fixed before any model

ORIENTATION

0.71 → 0.99

agreement before → after the flip

The building maps were stored upside down relative to the wind fields. One vertical flip aligns them.

OPEN-GROUND MASK

59 cases

mean wind changed by more than 1%

Some files hold values inside buildings, and some open cells hold exactly zero. Open ground now comes from the map (H = 0), without the zeros. Largest change: 14.6%.

WIND SCALE

4.5%

spread of u* between cases

Each case has its own friction velocity u*, from 0.2015 to 0.2106 m/s. Every case is divided by its own value.

One shared loader applies every fix, so all later stages read the same corrected data.

3 cases excluded: their downloadable map does not match the simulated domain. 6 kept but flagged; removing them changes R^2 by at most 0.007.

What we predict: the mean wind, in units of u*

1
Wind field
time-averaged speed U at 1.75 m

→

2
Keep open ground
cells where the building height H = 0

→

3
Average
over all open-ground cells

→

4
Divide by u*
the case's own friction velocity

→

TARGET
`U_mean / u*`
no units, so cases are comparable

$$U_{mean} = \frac{1}{|\Omega|} \sum U_i / u^*$$

Ω = the open-ground cells of one case

What is u*?

The friction velocity: the wind scale set by the drag of the ground and the buildings, here 0.2015 to 0.2106 m/s. It follows from the geometry, so dividing by it uses no simulated wind.

Other targets

U_p95: the speed exceeded on 5% of open ground. U_std and mean TKE: how much the wind varies. Task B uses U / u* at every open cell.

S03: test only on cities the model has never seen

535 USABLE CASES

482 development cases
93 groups: used for every training and tuning choice

53 test
24 groups

Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Frozen,
97 cases	97 cases	97 cases	96 cases	95 cases	used in S09

What is a group?

One city with all its neighbourhoods, or one family of idealized layouts with both wind directions and both height spreads. A group is never split across folds, so every fold holds out whole cities.

The folds, version 2

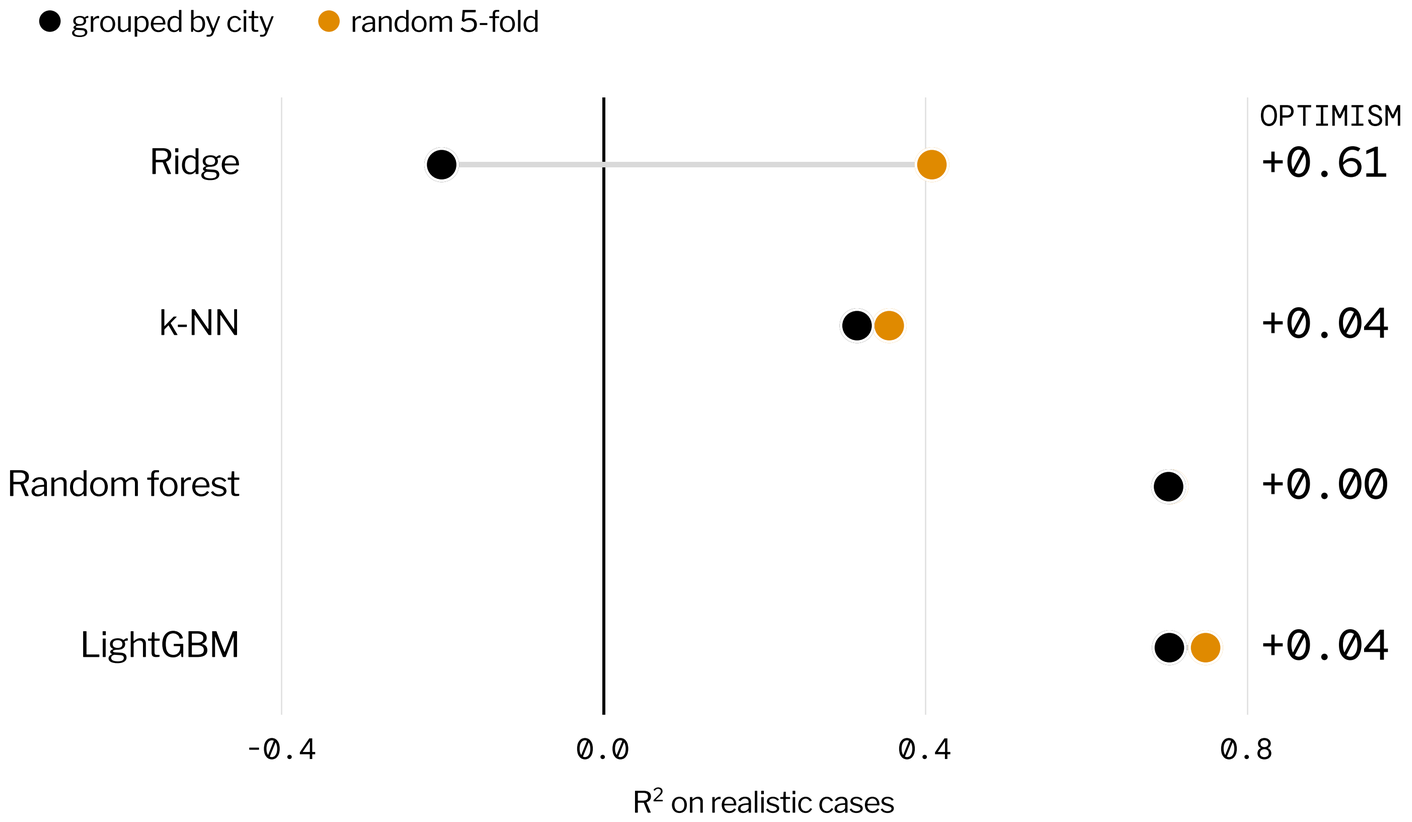
Version 1 grouped cases by name prefix, which split 7 cities, such as Melbourne, across several groups. Version 2 merged them: 136 groups became 93. The test groups were never changed.

The test set was fixed before any modelling and is used only in the final stage, S09.

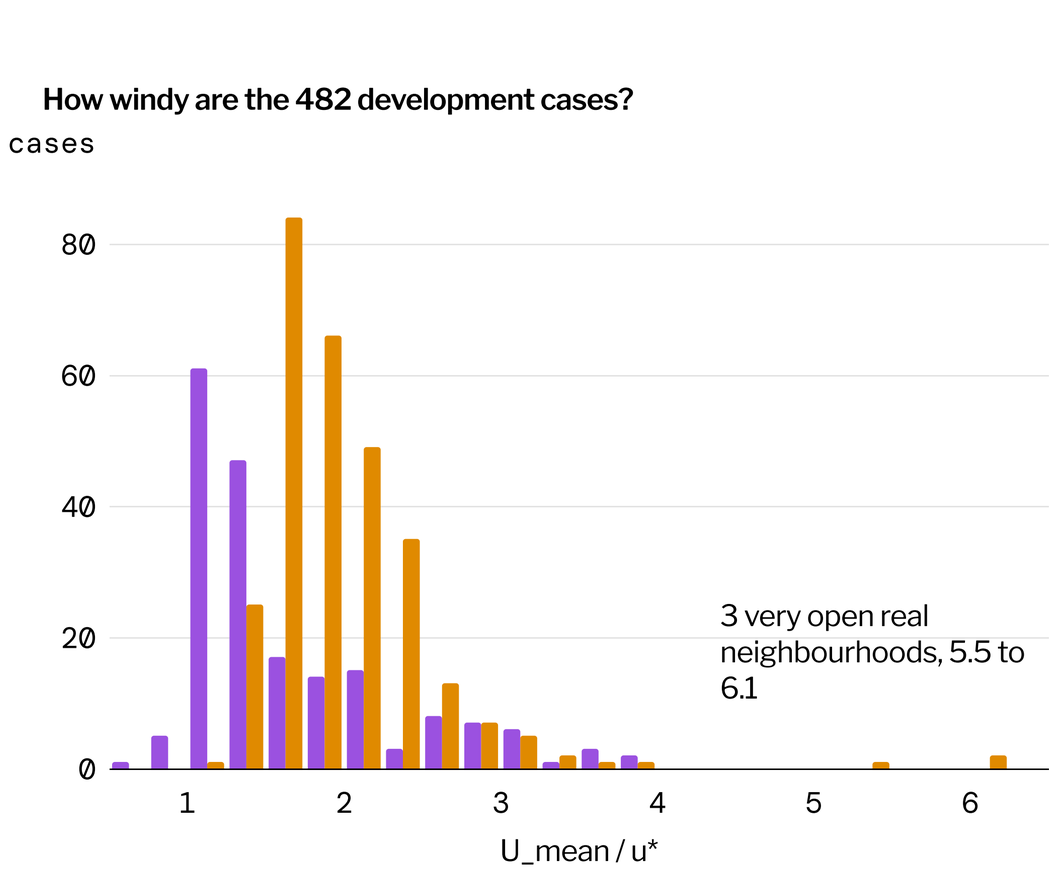
Why groups: random splits flatter the scores

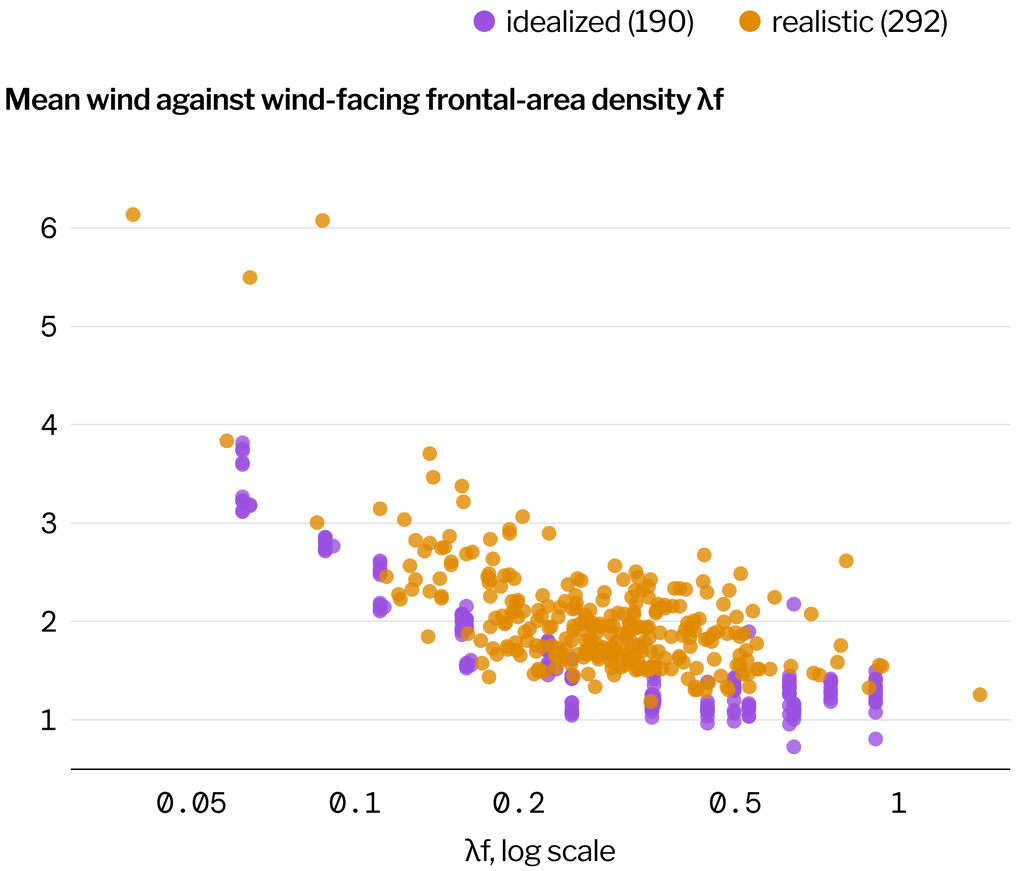
Neighbourhoods of one city look alike. If they sit on both sides of a split, the test partly measures memory, not generalisation.

Same untuned models and features, realistic cases only. Only the split changes.



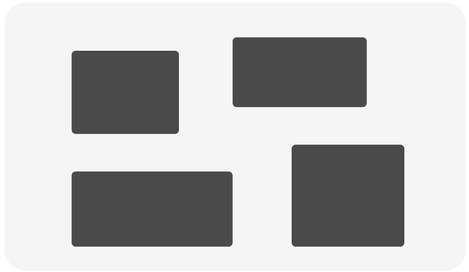
S04: more frontal area, less wind





Idealized mean 1.64, realistic 2.00. Arrays form tight clusters, but real neighbourhoods scatter widely at the same density: density alone is not enough.

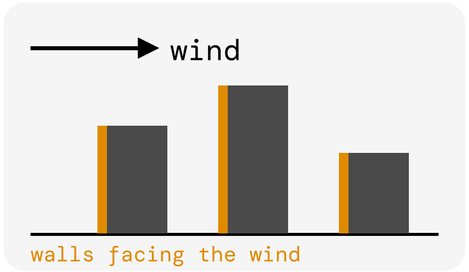
S05: describing a neighbourhood with physics



Plan-area density λp

building footprint ÷ ground area

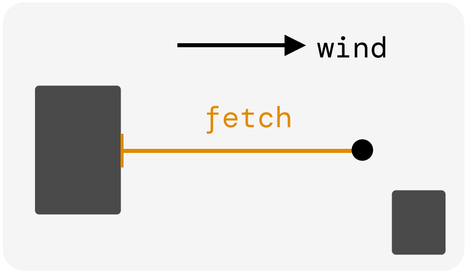
how much of the ground is built on



Frontal-area density λf

wall area facing the wind ÷ ground area

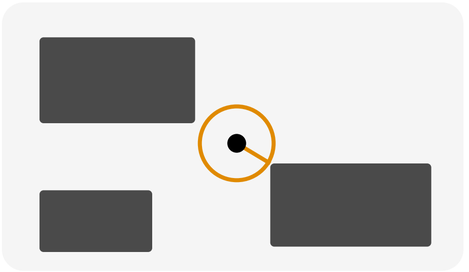
how much drag the wind meets



Upwind fetch

open distance upwind of a point, to the first building

how exposed a spot is to the wind



Distance to the nearest wall

radius of the open space around a point

how open a spot is in every direction

Roughness from published formulas

Macdonald (1998) and Kanda (2013) turn densities and heights into a roughness length z0 and a displacement height d: how strongly the city slows the wind above it.

INPUTS: BUILDING MAP AND WIND DIRECTION ONLY

THE DOMAIN REPEATS: DISTANCES WRAP AROUND THE EDGES

34 numbers per case, 30 per open cell

CASE LEVEL: ONE ROW PER SIMULATION

Basic 9

λp; mean, max and spread of height; λf(θ); wind direction (θ, cos θ, sin θ); subset flag

Morphology +8

wall-area density; height skewness and variation; built volume per area; buildings per hectare; mean footprint

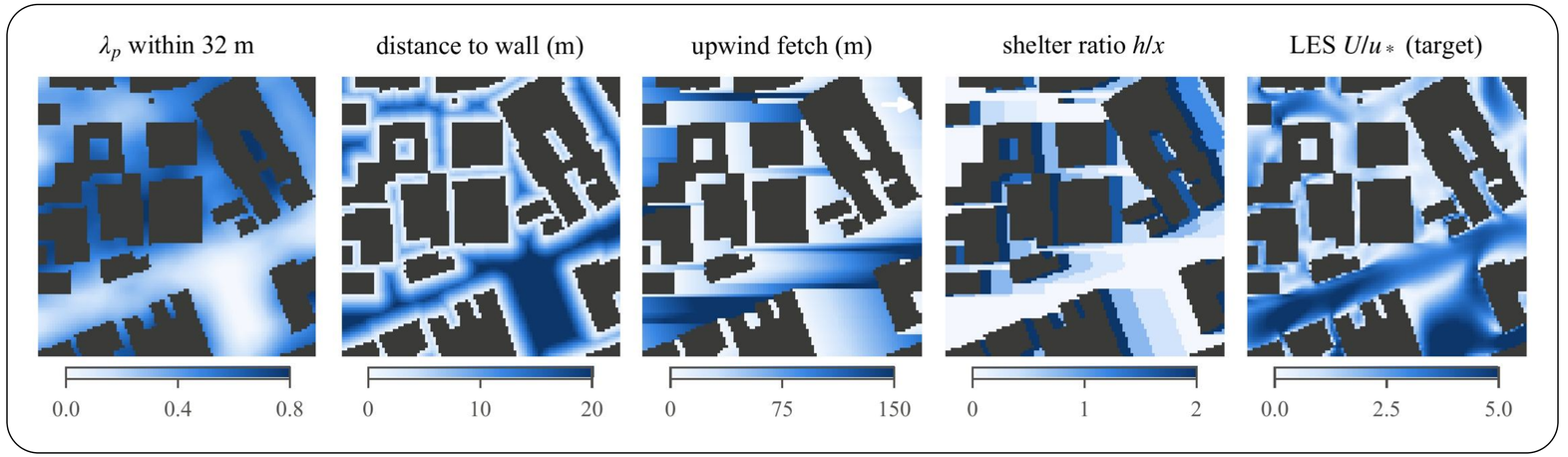
Physics +17

λf along x and y; Macdonald and Kanda d and z0; upwind, downwind and lateral fetch; distance to the nearest wall; facet orientation; windward wall share

PIXEL LEVEL: ONE ROW PER OPEN CELL

30 features per cell

local density and height within 8 to 64 m; distance to the nearest wall; fetch in four directions; tallest building upwind in three distance bands; canyon width and aspect ratio; wind direction; 6 case descriptors; 4 sheltering features (h / x)



Part of a realistic case (Adelaide), wind towards +x: four per-cell features and the simulated target.

S06: eight models, from simple to flexible

REFERENCE

Training mean

Always predicts the training average: the floor every model must beat.

LINEAR

Ridge

A weighted sum of the features, with a penalty that keeps the weights small.

SIMILARITY

k-NN

Averages the wind of the k most similar training cases.

KERNEL

SVR (RBF)

Fits a smooth curve and ignores errors smaller than a margin ε.

KERNEL

Gaussian process

Smooth interpolation between known cases, with one length scale per feature.

BAGGING

Random forest

500 trees on resampled cases and random feature subsets; their average cancels noise.

BOOSTING

XGBoost

Adds trees one at a time, each fixing the errors so far; grows them level by level.

BOOSTING

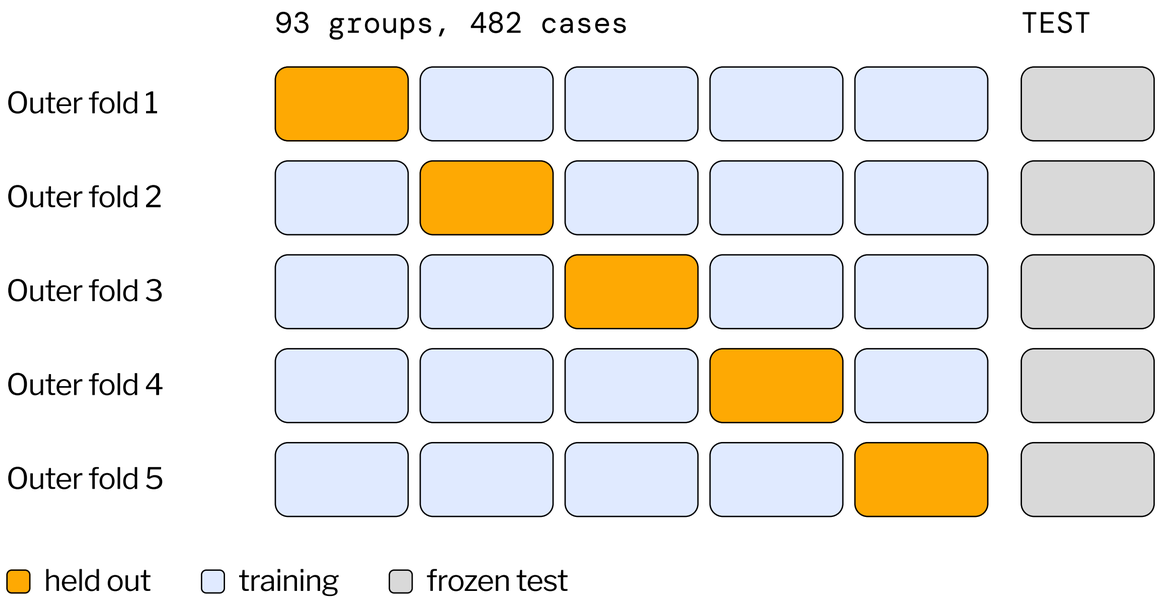
LightGBM

The same boosting idea, but each tree grows leaf by leaf, where it helps most.

SIMPLE ▶ FLEXIBLE

Baseline B1: a random forest with default settings on the 9 basic features, close to the inputs of earlier published work.

Tune inside the folds, score outside them



Each row is one outer round. The orange fold is scored; the four blue folds are used for training and tuning only.

1 Hold out one fold
whole cities and layout families that the model never sees

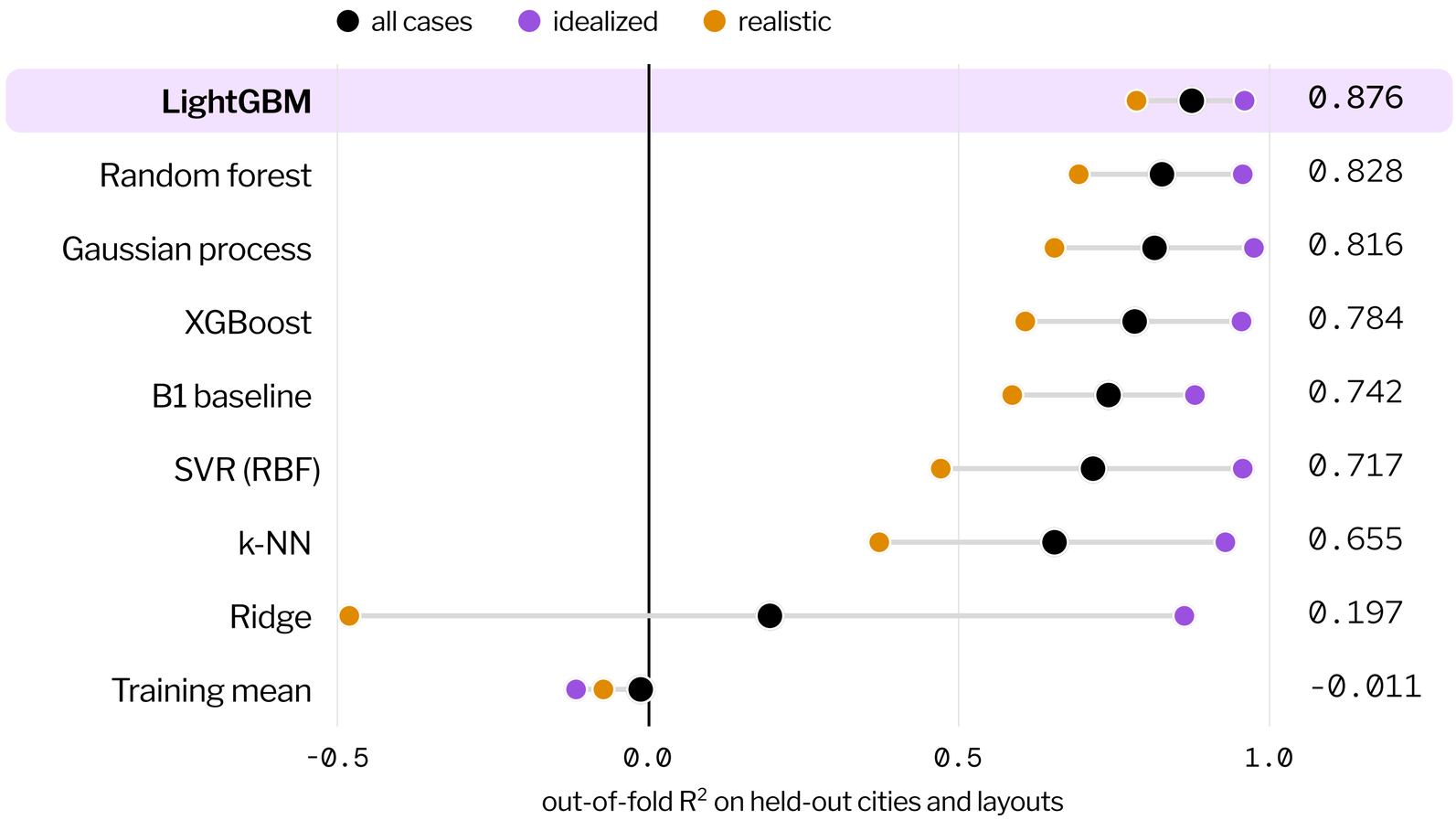
2 Tune on the other four
an inner 5-fold grouped search: Optuna, 30 trials

3 Refit, then score once
refit on all four folds, then scored on the held-out fold

4 Repeat 5 times and pool
every case gets exactly one out-of-fold prediction

Scaling and target standardisation are fitted inside the loops. Seed 42 throughout.

LightGBM predicts the mean wind best: R^2 0.876



How to read it

R^2 is the share of the case-to-case variation a model explains: 1 is perfect, 0 is no better than the average.

Real cities decide the ranking

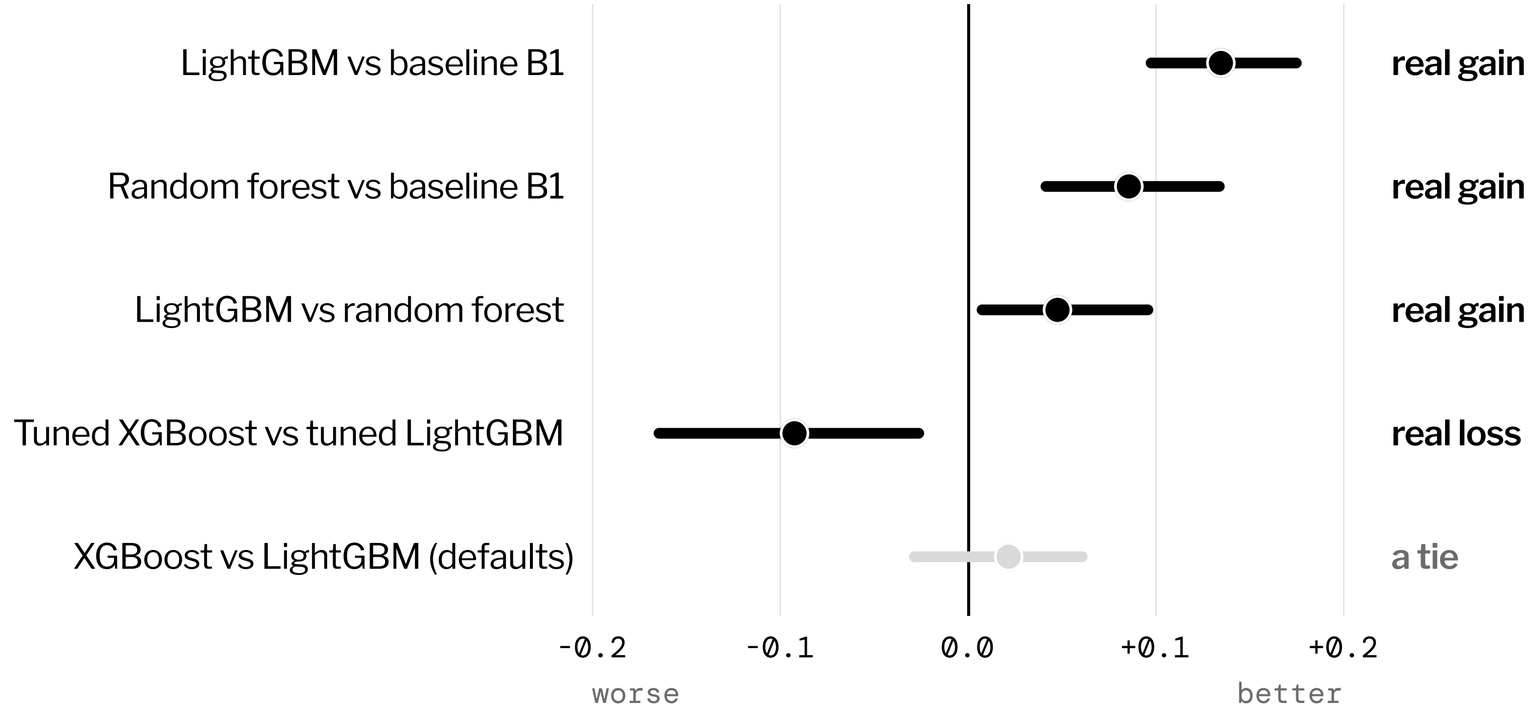
Idealized arrays are easy for almost every model. On realistic cases LightGBM reaches 0.787.

Typical error

LightGBM MAE 0.143 u*, about 8% of the mean wind.

Is the lead real? Yes, against the baseline

Gain in R^2 (dot), with its 95% range (bar)



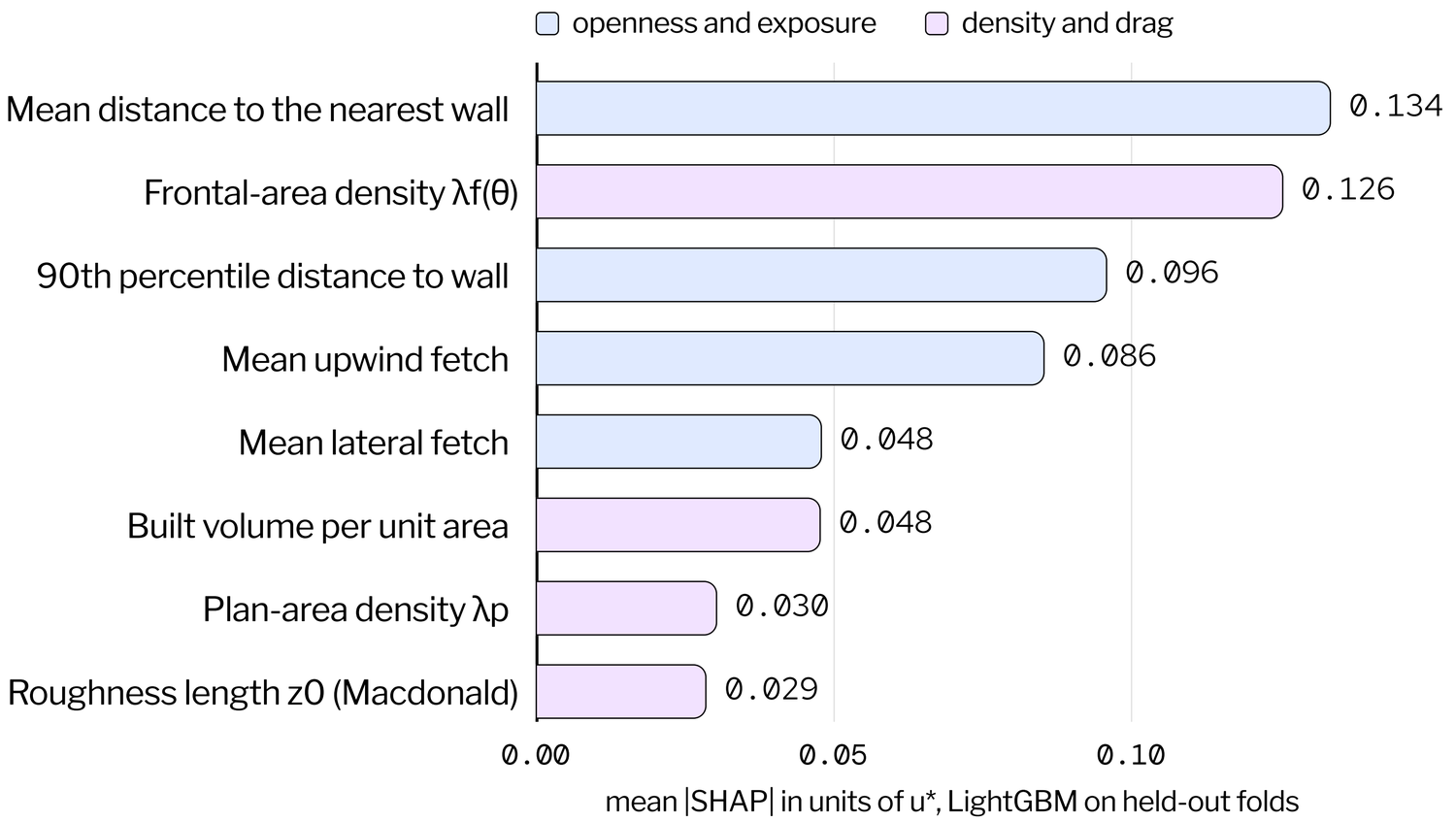
How we checked

We reshuffled the 93 city groups 2,000 times and measured each gap again. If the bar stays on one side of zero, the gap is real.

Why tuned XGBoost lost

In one fold, its tuning chose a setting that averages rare cases away. It underpredicted two very open neighbourhoods. With default settings the two models tie.

What drives the wind: openness and frontal area



What SHAP does

It splits each prediction into one contribution per feature. The bars average their size over held-out cases.

The signs make physical sense

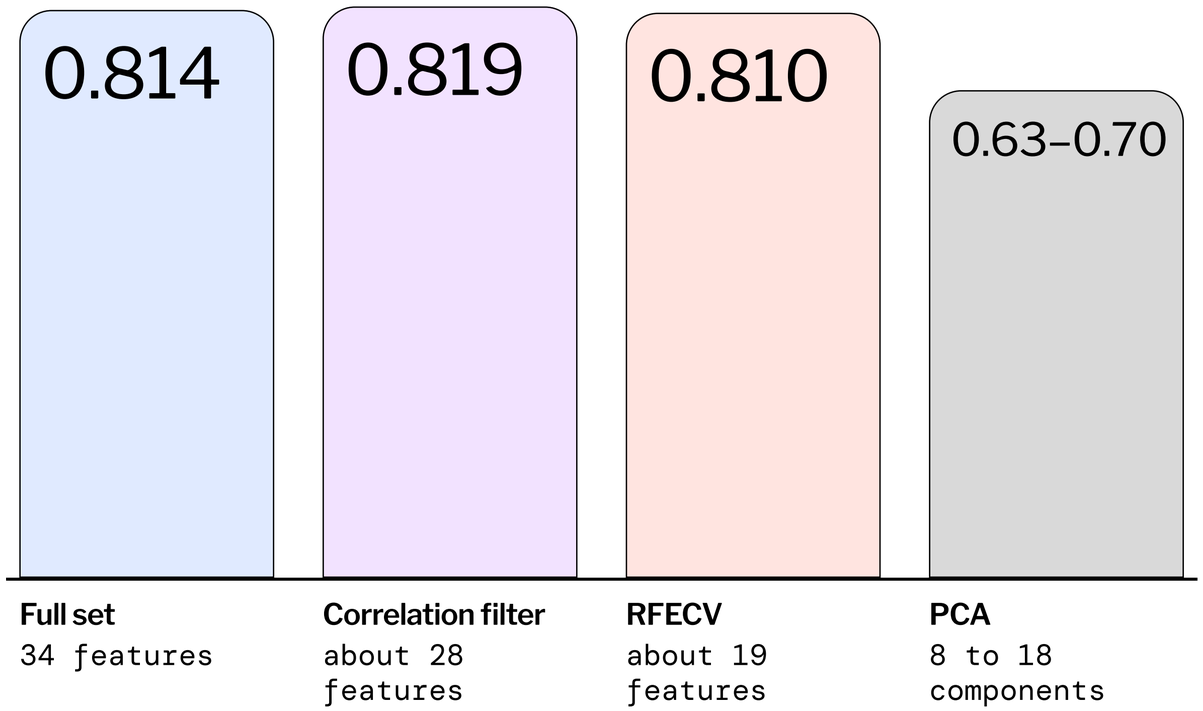
More open space and less frontal area raise the mean wind, in both subsets.

Caveat

Correlated features share credit, so this ranks how the model uses them, not causes.

S06b: fewer features did not help, and PCA hurt

pooled out-of-fold R^2, LightGBM with fixed settings



Setup

Selection, scaling and PCA are fitted inside each training fold only; the held-out fold is never seen.

No clear gain

The first three scores are within 0.01 of each other.

Why PCA likely hurts

It mixes the features into combined axes, which blurs the clean physical thresholds that trees split on.

Decision: keep all 34 features.

S07: from one number to the whole wind map

Where does the wind vary? Share of all variation at street level

31%	69%
between neighbourhoods	within each neighbourhood: jets, wakes, corners

A flat map at the true mean wind scores only R^2 0.311. So getting the average right is not enough: we need the pattern inside each map.

The model

LightGBM on the 30 features of every open cell, with fixed settings.

Training

3,000 random cells from each training case, so every case counts equally.

The hybrid

Predict the pattern, then multiply it by the case-level mean from S06.

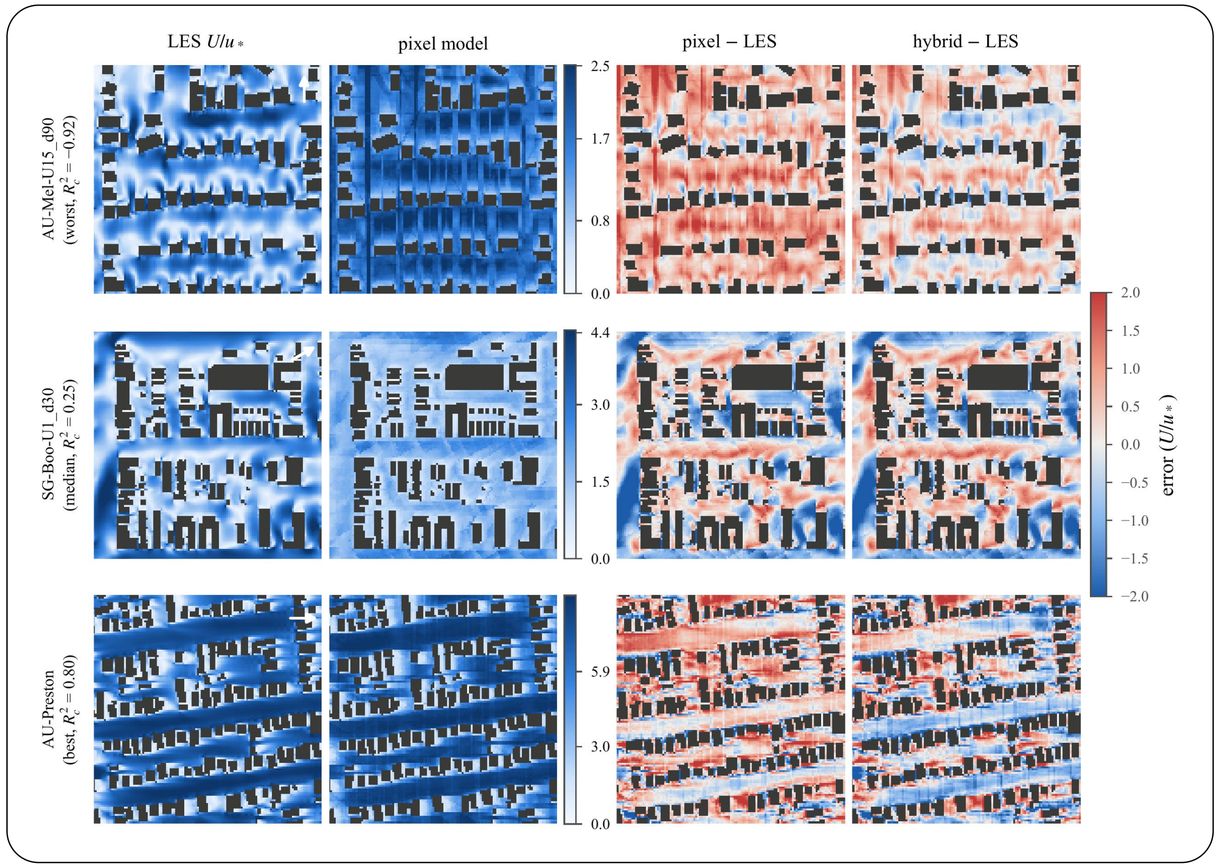
Pooled R^2

all cells of all maps together: rewards getting each map's level right

Per-case R^2

inside one map only: measures the pattern

Right level, smoothed pattern



Unseen real neighbourhoods. Columns: simulation, prediction, and two error maps (blue too slow, red too fast).

0.591

pooled R^2 (a flat map at the true mean: 0.311)

0.42 / 0.25

per-case R^2, idealized / real neighbourhoods

Peaks too low

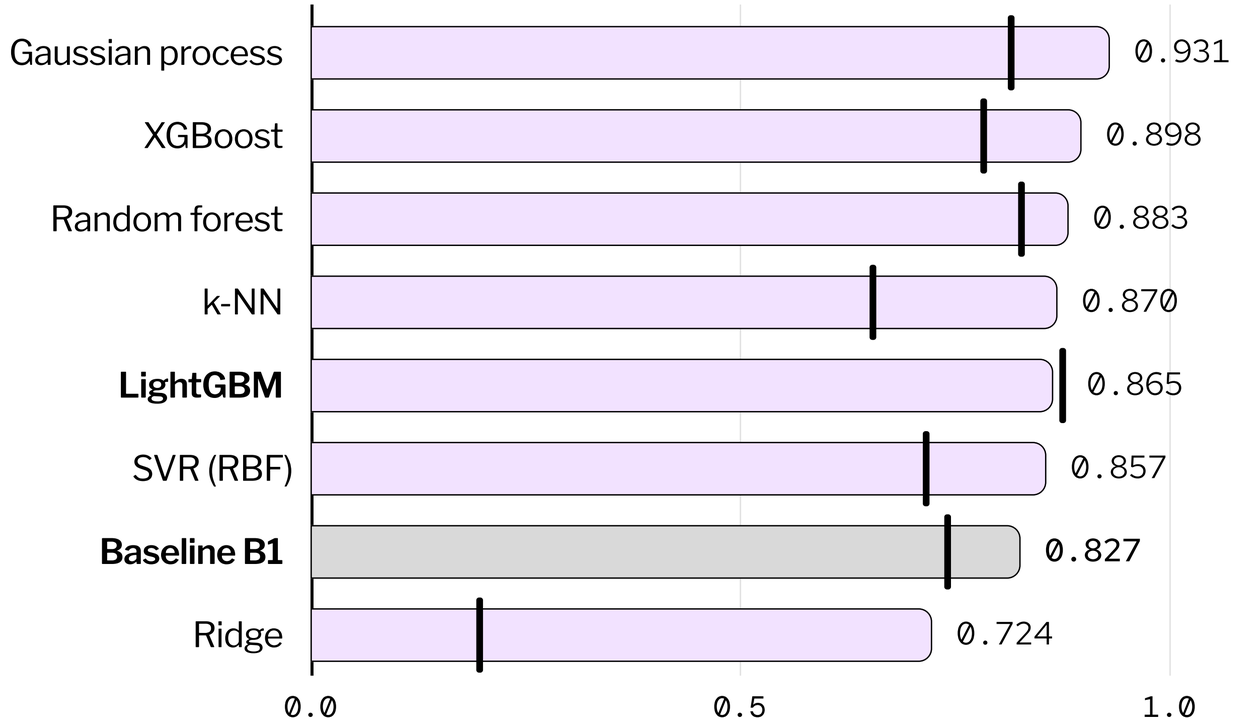
the fastest 5% of wind is underpredicted by 0.73 u*

The limit is the features

A random split or a much bigger model barely helps on unseen cases. Seeing jets and wakes needs a model that reads the map itself.

S09: the frozen test set confirms it generalises

R^2 for the mean wind. Bars: test (53 cases). Black ticks: development (482 cases).



Scored only at the end

53 cases in 24 groups (34 idealized, 19 real), scored only after every choice was fixed.

Non-linear models beat the baseline

Every one of them scores above B1 (0.827).

The wind map holds up

Pooled R^2 0.680, against 0.378 for a flat map at the true mean.

Why the GP tops this chart

The test has only 19 real cases and is mostly block arrays, where the GP is strongest. That is too few cases to rank close models. The larger development comparison (black ticks) ranks LightGBM first, and we fixed that choice before seeing the test.

What Phase 1 showed

1. Clean data first: three fixes changed the targets before any model was trained.
2. From geometry alone, LightGBM predicts a neighbourhood's mean wind with R^2 0.876 on unseen cities.
3. Open space and frontal area drive the wind, as the physics suggests.
4. The wind map gets each neighbourhood's level right but smooths its jets and wakes.

Limits: simulated neutral wind only, no trees or terrain, one height (1.75 m), and only 19 real test cases.

Phase 2

Read the building map directly

A U-Net, a convolutional network that reads the whole building map, with wrap-around padding.

A hybrid: the case-level mean from Phase 1 times a learned pattern.

Deep ensembles, to give every prediction an uncertainty.

The same grouped folds, so success still means unseen cities.

TEAM T070

Thank you

WHO DID WHAT

Mahir Abdullah	Designed and coordinated the pipeline; audited all 538 cases and built the shared loader and evaluation tools; case-level model comparison, wind-map model, final test and the report.
Neeraj Singh	Exploratory analysis of the profiles, building maps, wind fields, forcing and time series; data-quality checks; the master case table and the model-ready case list.
Om Mishra	Preprocessing of the vertical profiles; the grouped frozen splits, the evaluation metrics and the evaluator.
Aparna Singh	The physics-informed feature library and its tests; the first nested tuning of the case models; the study of feature selection and PCA.
Ayush Aryan	The first case-level model ladder, from linear models to boosted trees, and the feature-set ablation.

Code and report: github.com/mahir-m01/Urban-Microclimate-Modelling. Data: UrbanTALES (Nazarian et al., 2025), CC BY-NC 4.0.